

Aim: Inheritance, Overloading and Overriding.

1. Write a java program that demonstrate the usage of Inheritance by using suitable class: A class called Inher_1 that defines an array containing numbers of int type. A class called Inher_2 inherits Inher_1. It defines a method called menu(k) to display menu as numbers from Inher_1, sum of numbers, average of numbers and maximum numbers. A class InherDemo defines main () method and an object defined for Inher_2. Invoke menu (k) method to get appropriate results by passing different values of k.

CODE:

```
import java.util.Scanner;
class Inher1{
    int x[]={1,2,3,4,5};
}
class Inher2 extends Inher1{
    void menu(int k){
        int sum = 0;
        for(int i=0; i<x.length; i++) {
            sum = sum+ x[i];
        }
        switch(k){
            case 1:
                System.out.println("Your entered elements are: ");
                for(int i=0; i<x.length; i++) {
                    System.out.print(x[i] + " ");
                }
                break;
            case 2:
                System.out.println("The sum of all the elements is: " + sum);
                break;
            case 3:
                System.out.println("The average of the elements is: " + (double)sum/x.length);
                break;
            case 4:
                int max = x[0];
                for (int i = 1; i < x.length; i++) {
                    if (x[i] > max) {
                        max = x[i];
                    }
                }
                System.out.println("The maximum element is: " + max);
                break;
            default:
                System.out.println("Invalid option");
        }
    }
}
public class PracFourOne{
    public static void main(String[] args) {
        Inher2 obj = new Inher2();
```

```
Scanner sc = new java.util.Scanner(System.in);
System.out.println("Enter your choice: ");
System.out.println("1. Display elements");
System.out.println("2. Sum of elements");
System.out.println("3. Average of elements");
System.out.println("4. Maximum element");
int choice = sc.nextInt();
obj.menu(choice);
sc.close();
}
}
```

OUTPUT:

Enter your choice:

1. Display elements

2. Sum of elements

3. Average of elements

4. Maximum element

4

The maximum element is: 5

2. Write a java program that demonstrate working of Super keyword.**CODE:**

```
class Base{
    int n1,n2;
    Base(int x,int y){
        n1=x;
        n2=y;
    }
    int sum(){
        return n1+n2;
    }
}
class Child extends Base{
    int max;
    Child(int x,int y){
        super(x,y);
    }
    void show(){
        if(n1>n2)
            max=n1;
        else
            max=n2;
        System.out.println("Maximum number:"+max);
        System.out.println("Addition is "+sum());
    }
}
class SuperDemo{
    public static void main(String args[]){
        Child c = new Child(20,90);
    }
}
```

```
    c.show();  
}  
}
```

OUTPUT:

Maximum number:90

Addition is 110

3. Write a java program that illustrate the concept of Method Overloading.**CODE:**

```
class Box{  
    double ht,wt;  
    Box(){  
        ht=10;  
        wt=10;  
    }  
    Box(double x){  
        ht=x;  
        wt=x*x;  
    }  
    Box(double a,double b){  
        ht=a;  
        wt=b;  
    }  
    Box(Box bx){  
        ht=bx.ht;  
        wt=bx.wt;  
    }  
    void display(){  
        System.out.println("The box has height:"+ht+"and weight:"+wt);  
    }  
}  
  
public class Overloading{  
    public static void main(String args[]){  
        Box b = new Box();  
        Box b1 = new Box(9);  
        Box b2 = new Box(4,6);  
        Box b3 = new Box(b);  
        b.display();  
        b1.display();  
        b2.display();  
        b3.display();  
    }  
}
```

OUTPUT:

The box has height:10.0and weight:10.0

The box has height:9.0and weight:81.0

The box has height:4.0and weight:6.0

The box has height:10.0and weight:10.0

4. Write a java program that illustrate the concept of Constructor Overloading.**CODE:**

```
import java.util.Scanner;
class Sh{
    int area(int s){
        return s*s;
    }
    double area(double x,double y){
        return x*y;
    }
}
public class MOverloading {
    public static void main(String args[]) throws Exception{
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter side of square:");
        int s = sc.nextInt();
        Sh sp = new Sh();
        System.out.println("Area of Square:"+sp.area(s));
        System.out.println("Area of Rectangle:"+sp.area(10.0,20.0));
        sc.close();
    }
}
```

OUTPUT:

```
Enter side of square:15
Area of Square:225
Area of Rectangle:200.0
```

5. Write a java program that demonstrate the usage of Method Overriding.**CODE:**

```
class Parent1{
    void display(){
        System.out.println("I am from parent class!");
    }
}
class Child1 extends Parent1{
    void display(){
        System.out.println("I am from child class!");
    }
}
public class Overriding {
    public static void main(String args[]){
        Child1 c = new Child1();
        c.display();
    }
}
```

OUTPUT:

```
I am from child class!
```